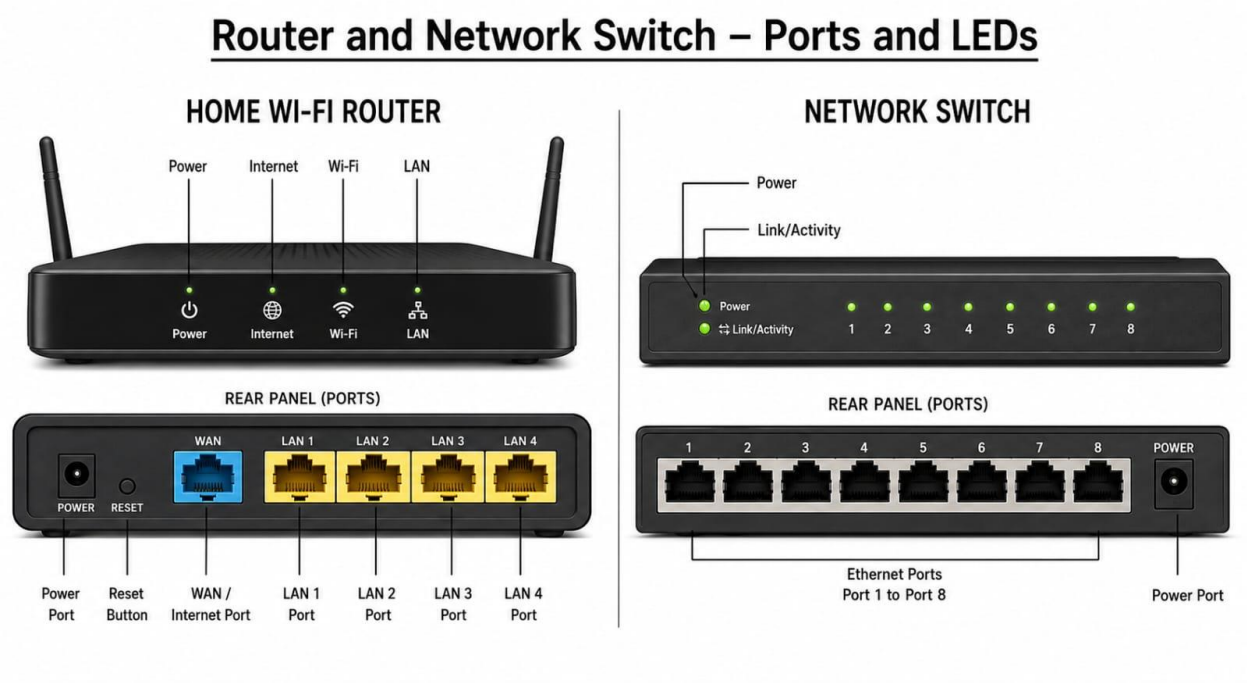
Q1.Draw a labelled diagram of a home Wi-Fi router and a network switch.



Labels and their functions:

- **Power Port:** Supplies electricity to the device.
- **WAN Port:** Connects the router to the modem or internet connection.
- **LAN Ports:** Connect computers and other wired devices.
- **Console Port:** Used to configure some routers and managed switches.

- **Status LEDs:** Show power, internet connection, Wi-Fi, and network activity.

- **Switch Ethernet Ports:** Connect multiple wired devices in a LAN.

Q2. Write 3 key differences between routers and switches.

No.	Router	Switch
1. Main function	Connects different networks.	Connects devices within a LAN.
2. Address used	Uses IP addresses to route packets.	Uses MAC addresses to forward frames.
2. Traffic handling	Sends data between different networks.	Sends data to the appropriate device within a LAN.

Q3. Identify ports and LEDs on a router and switch.

A. Router

1. Power Port: Located on the rear side; connects the power adapter.

2. WAN Port: Located on the rear side; connects to the modem or internet source.

3. LAN Ports: Located on the rear side; connect computers using Ethernet cables.

4. Power LED: Usually on the front; indicates that the router is powered on.

5. Wi-Fi LED: Indicates wireless network activity.

6. Internet LED: Indicates the internet connection status.

B. Network Switch

1. Power Port: Supplies power to the switch.

2. Ethernet Ports: Connect computers, printers, and other wired devices.

3. Power LED: Indicates that the switch is switched on.

4. Link/Activity LEDs: Show whether a port has a connection and is transferring data.

Practical activity: Use the diagram in Q1 to identify these parts. If your teacher requires a real image, take a photo of your own router and switch and label the visible ports and LEDs.

Q4. Gaming Café Network Scenario

I would first connect the ISP internet line to the modem and connect the modem to the router's WAN port. Then, I would connect one of the router's LAN ports to a network switch using an Ethernet cable.

Next, I would connect all 8 gaming PCs to the switch using Ethernet cables. The router would manage IP addresses and route internet traffic, while the switch would connect all gaming PCs within the local network.

Justification: A router is needed to connect the café network to the internet, and a switch provides enough Ethernet ports for all 8 gaming PCs. A suitable gigabit switch would support fast, stable connections for online gaming and IPL streaming.